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BMJ Health & Care Informatics

Perspectives on telemedicine across urban, rural and remote areas in the Philippines during the COVID-19 pandemic

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ABSTRACT

Objectives This study explored attitudes, subjective norms, and perceived behavioural control of participants across urban, rural and remote settings and examined intention-to-use telemedicine (defined in this study as remote patient–clinician consultations) during the COVID-19 pandemic.

Methods This is a cross-sectional study. 12 focus group discussions were conducted with 60 diverse telemedicine user and non-user participants across 3 study settings. Analysis of responses was done to understand the attitudes, norms and perceived behavioural control of participants. This explored the relationship between the aforementioned factors and intention to use.

Results Both users and non-users of telemedicine relayed that the benefits of telemedicine include protection from COVID-19 exposure, decreased out-of-pocket expenses and better work–life balance. Both groups also relayed perceived barriers to telemedicine. Users from the urban site relayed that the lack of preferred physicians discouraged use. Users from the rural and remote sites were concerned about spending on resources (ie, compatible smartphones) to access telemedicine. Non-users from all three sites mentioned that they would not try telemedicine if they felt overwhelmed prior to access.

Discussion First-hand experiences, peer promotions, and maximising resource support instil hope that telemedicine can help people gain more access to healthcare. However, utilisation will remain low if patients feel overwhelmed by the behavioural modifications and material resources needed to access telemedicine. Boosting infrastructure must come with improving confidence and trust among people.

Conclusion Sustainable access beyond the pandemic requires an understanding of factors that prevent usage. Sufficient investment in infrastructure and other related resources is needed if telemedicine will be used to address inequities in healthcare access, especially in rural and remote areas.

WHAT IS ALREADY KNOWN ON THIS TOPIC.

⇒ Telemedicine services allow healthcare providers to reach patients beyond healthcare centres through phone, social media, and other remote platforms.

WHAT THIS STUDY ADDS

⇒ This study provides a comparative analysis of the personal and infrastructural factors that may affect telemedicine utilisation among urban, rural and remote localities in low to middle income countries (LMICs). It delivers powerful grassroots-level narratives on how telemedicine can become people-centred during the pandemic and beyond.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ This study provides an in-depth analysis of how policy-makers can maximise resources to sustain utilisation of telemedicine beyond the pandemic and in anticipation of the universal healthcare implementation in the Philippines. It provides a guide on how primary healthcare advocates can improve technological infrastructure while making people confident about integrating technology into their healthcare routines.

INTRODUCTION

Background

Telemedicine has shown potential in reducing poor health outcomes and healthcare-related expenditures.[1 2] Remote consultations in the USA allowed healthcare providers to give necessary prescriptions and provide immediate clarification for health-related queries, decreasing the needed frequency of in-person consultations.[3] A 2015 study in Canada reported that 85% of patients expressed intent of continuing telemedicine use due to an average cost reduction of US$200 for them and their caretakers.[4]

The impact of telemedicine varies by setting and availability of resources. A study in Southeast Asia reported limited access to gadgets and internet sources to sustain telemedicine use in certain countries such as Indonesia and the Philippines.[5] To expand internet access in the latter, Executive Order (EO) 127 was passed. This EO aimed to address the

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gap in telemedicine care through inclusive access to satellite services.[6] However, the distance of signal sites from communities remained a barrier to obtaining healthcare, with 57.4% of Filipinos remaining offline.[7] Technology-related anxiety also deters patients and caretakers from investing in telemedicine care.[8 9] Furthermore, training healthcare workers to navigate the platform has yet to be integrated into healthcare training curricula.[5]

The COVID-19 pandemic hastened the development of alternative means for utilisation of primary care services.[10–12] The Philippine Primary Care Studies (PPCS) introduced telemedicine services to sustain primary care service utilisation during the pandemic and beyond. The PPCS team expanded internet access in its three partner sites to provide telemedicine services. The initial implementation of telemedicine services was launched on platforms that participants had easy access to. In the context of this study, telemedicine refers to remote patient–clinician consultations. The telemedicine platforms made available in the study sites were Facebook Messenger (with consultations done through video call, phone call and chat), and phone call using a landline phone for patients who had difficult access to the internet despite the provision of Wi-Fi access in the sites.

Telemedicine's potential cannot be fully maximised by patients without uncovering the factors that either increase hesitancy or fully hinder utilisation. Thus, this study aimed to explore attitudes, subjective norms and perceived behavioural control of telemedicine users and non-users in urban, rural and remote settings and to examine intention-to-use telemedicine during the COVID-19 pandemic.

METHODOLOGY

Study design

This is a cross-sectional study on factors that affect telemedicine usage. Data gathering was conducted from February to March 2022 through focus group discussions (FGDs) to assess experiences and attitudes towards telemedicine as a mode of healthcare delivery among end-users at the height of the COVID-19 pandemic. FGDs were used for data collection. This allowed the team to understand shared and diverging perspectives from groups who have common backgrounds. FGDs were done online due to the to prevent COVID-19 transmission during the pandemic.

A semistructured interview guide (online supplemental appendix A) was adapted from George *et al*.[13] We conducted a pilot test of this interview guide with patients from the urban, rural and remote sites. Based on the results of the pilot test, we added questions on prior knowledge about telemedicine, preferred platforms of use, and desired features. We also added questions about the necessity to consult during a pandemic to assess how COVID-19 may have influenced telemedicine utilisation behaviour.

Table 1 Demographic characteristics of 60 study participants

	Urban site n (%)	Rural n (%)	Remote n (%)	Total n (%)
Sex				
Female	14 (70)	19 (95)	17 (85)	50 (83)
Male	6 (30)	1 (5)	3 (15)	10 (17)
Residence				
Near health facility	10 (50)	10 (50)	10 (50)	30 (50)
Far from health facility	10 (50)	10 (50)	10 (50)	30 (50)
Telemedicine use				
User	10 (50)	10 (50)	10 (50)	30 (50)
Non-user	10 (50)	10 (50)	10 (50)	30 (50)

Participants and study site

We included patients who consulted from April 2020 to March 2022 in the study. Recruitment was done in a reverse sequence order. The team first contacted the patients who consulted most recently to secure consent for participation.

UPHS[14] was the urban site of the study. It was a 25-bed facility serving a population of approximately 31 000 students, faculties, employees and campus residents. 12 family physicians were employed during the study period.

Samal, Bataan[14] was a fourth-class rural municipality with approximately 36 000 residents during the study period. Residents were able to avail of primary care services through the rural health unit (RHU) or barangay health stations (BHSs). The RHU hosted doctors, nurses and midwives. BHSs were primarily supervised by nurses and midwives.

Bulusan, Sorsogon[14] was a fourth-class municipality with approximately 23 000 residents. It was declared as a geographically isolated and disadvantaged area by the Philippine Department of Health. This classification can be attributed to its mountainous terrain, unstable connectivity, and limited transportation options. Residents may also receive care from either RHU or BHS. Both rural and remote areas only had one physician at the start of the study. Purposive sampling was done. Demographic distribution can be seen in Table 1.

Four FGDs per site were conducted. Two FGDs included participants who resided near the health facilities (on-campus residents for the urban site and people living in the lowland barangays for the rural and remote sites): one FGD included participants who were users of telemedicine while the other FGD included participants who were non-users. Telemedicine usage may vary over time. In this study, we defined 'users' of telemedicine as individuals who have used telemedicine through any platform at the time of the study, and 'non-users' as individuals who have not used telemedicine using any platforms at the time of the study.

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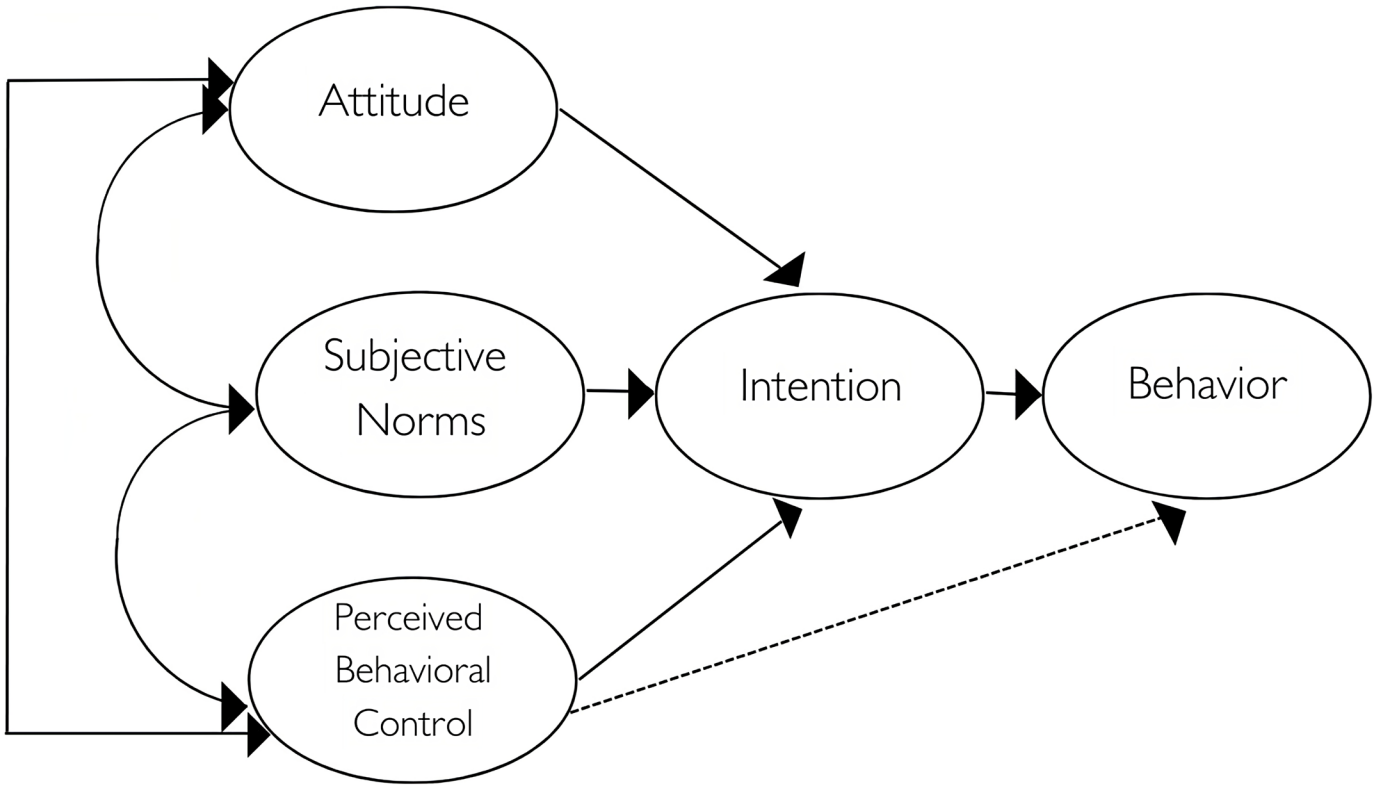


Figure 1 Diagram of theory of planned behaviour framework.

Thematic analysis

As a framework, the theory of planned behaviour examines how attitudes, subjective norms and perceived behavioural control impact the intention of a person to adopt a particular behaviour (Figure 1).[15] We analysed the data and classified responses according to domains, including internal perceptions (referring to attitudes), external influences (referring to subjective norms) and availability of resources (referring to perceived behavioural control). Categorising data into these domains allowed us to compare which domains strongly informed telemedicine utilisation behaviour among participants. Data were processed using NVivo V.12. The research lead performed the initial coding. Elicited themes reveal participant perceptions and/or experiences under each domain. Codes were then presented to a multidisciplinary team of public health practitioners to determine if codes aligned with sentiments elicited from the study. Dissenting opinions on the themes were resolved by reviewing the transcript to verify the context of a participant's response.

RESULTS

Key demographics

A total of 60 participants were included in this study. There were more females (83%, n=50) in our study population.

Response of telemedicine users

A comprehensive overview of sentiments from participating telemedicine users is found on Table 2. Under the attitude domain, telemedicine was largely perceived to increase the respondents' feeling of safety against COVID-19. Motivators for telemedicine use included its applicability for minor and chronic illnesses. Respondents also espoused that it helped them save time, money and effort travelling to health facilities. However, deterring attributes made participants more hesitant towards continued use. A few participants shared that the specialisation of certain physician may not match the health concerns of a patients.

Exploration of the subjective norms domain revealed high awareness across all participants of our telemedicine platform's features. Information sources and prevailing sentiments from key persons highly influenced their perception. These included individuals whose opinions participants were most open to consider. Consultations via Facebook Messenger or call were considered straightforward. Participants were aware of how bookings are conducted, which was either through Facebook Messenger or call reservations in rural and remote sites and reservations through the online portal in the urban site. Awareness did not necessarily translate to maximisation of the online portal. Participants in the urban site would often maximise internal networks within the PPCS partner facility to book consults. CHWs were integral sources of information in the rural and remote sites. Through CHWs, they felt especially guided by a trusted source while exploring telemedicine.

Factors related to the perceived behavioural control domain had the greatest influence on the likelihood

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Table 2 Summarised responses of attitudes, subjective norms and perceived behavioural control among telemedicine users

Domain	Themes	Urban	Rural	Remote
Attitude	Motivating attributes	▶ Telemedicine prevents possible exposure to COVID-19 ▶ Telemedicine allows a patient balance work and health needs ▶ Telemedicine is particularly useful for minor injuries, obtaining fit to work permits, and monitoring chronic conditions ▶ Telemedicine allows consultation with a doctor who is already familiar with the patient's medical history	▶ Telemedicine prevents possible exposure to COVID-19 ▶ Telemedicine allows a patient balance work and health needs ▶ Increased certainty that consultations will push through ▶ Economical alternative ▶ Regard telemedicine as most fitting for minor and chronic ailments ▶ Recommended use for senior citizens and children	▶ Allows more time for agricultural work ▶ Appreciate that they do not have to travel long distances ▶ Economical alternative ▶ Regard telemedicine as most fitting for minor and chronic ailments ▶ Recommended use for senior citizens and persons with disabilities
	Deterring attributes	▶ Concerned about others feigning illness just to get a medical certificate since it makes the queue longer ▶ The perceived mismatch between providers' specialisation and patient's health concerns disincentivises use in the long-term	▶ Uncertain if telemedicine use is a financially wise decision. General perception that it might be more cost-effective to go to health centres.	▶ Uncertain if this telemedicine is a sustainable financial alternative. Primarily attributes this to the insufficiency of their income to purchase resources.
Subjective Norms	Source of information	▶ Information was mostly received through workd of mouth and announcements by the health facility	▶ Information was mostly provided by the Rural Health Unit Facebook page and Community Health Workers (CHWs)	▶ Information was mostly provided by the Rural Health Unit Facebook page and CHWs
	Influencing perceptions	▶ Majority tried telemedicine because it was required by their office ▶ Likely to consult online sources and peers for more information on a physician's credibility	▶ Positive feedback on telemedicine from neighbours	▶ Felt that feedback from neighbours influenced their decision to try the platform out
Perceived behavioural control	Reachability of providers	▶ Doctors had very limited availability as listed on the appointment setting platform, invoking a perception of inaccessibility despite infrastructure being present	▶ Agreeable to availabilities set by physicians who conducted online consultations	▶ Concerned about the sufficiency providers to address their concerns
	Familiarity	▶ Preferred intuitive platforms which may include the integration of telemedicine services in commonly used social media networks ▶ When technical difficulties arise, they feel like they are not able to maximise the appointment because time becomes even more limited	▶ Telemedicine services may be more accessible if it is installed in messenger or landline services	▶ Platforms must be intuitive, inclusive, and affordable
	Resource capacity	▶ Felt that they have the means to access telemedicine services when they want to. However, additional support will be greatly appreciated. This includes a more efficient system of streamlining appointments.	▶ Prefer platforms that are intuitive and affordable ▶ Inadequate material resources (ie, stable electricity supply, sufficient internet or mobile data connectivity) to access telemedicine services	▶ Lack of material resources (ie, phone) ▶ Increased susceptibility to calamities decreases behavioral control

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of continued telemedicine use. Urban site respondents generally perceived that they have good behavioural control as resource capacity and familiarity with the platform were high. High exposure to technological devices at their jobs gave them the confidence to use devices outside work. Respondents from the urban site relayed that technical glitches hinder the full utilisation of telemedicine services. One participant mentioned that 'it took long for my consultation request to get accommodated. I was passed on from one attendant to another before someone entertained my teleconsultation concern. I had to confirm with my doctor if the consultation request was approved'. Participants from the rural and remote sites were eager to continue using the service; however, the lack of material resources decreased the likelihood of consistent use. Participants in all sites agreed that integrating telemedicine in Facebook increased their level of confidence. A familiar platform allowed them to invest more effort on their health instead of learning the application.

Response of telemedicine non-users

A comprehensive overview of insights from participating non-users is found on Table 3. In the attitude domain, deterring factors more strongly influenced their decision to decline use. Non-users from the urban site emphasised that 'convenience was a case-to-case basis'. They relayed that the effectiveness of telemedicine was limited to people who were technologically competent and comfortable with discussing their concerns online. One participant also shared, 'I think the physician can assess me better when we see each other in-person. They are also more able to make me feel at peace in-person. Physical contact makes me more certain that an actual person is taking care of my health'.

In the subjective norms domain, non-users relayed that government officials or workplace personnel for the urban setting, and efforts of CHWs for the remote setting were integral sources of information. Hesitancy of the participants in the remote and rural sites to use telemedicine were related to their lack of familiarity with telemedicine and inadequate resources.

In the perceived behavioural control domain, low-resource capacity was the primary deterrent in both rural and remote sites. One participant mentioned that 'I was very eager to use telemedicine, but the cellular data signal is too weak in our area. I'm hesitant to invest a lot of money for load and an advanced smartphone since I must prioritize spending on my medications'. Participants also shared that investing in telemedicine-related resources is risky since natural calamities can cause power outages and decreased internet connectivity. All non-users in the urban site preferred face-to-face consultation. They felt that this avenue allowed them to consider alternative providers more carefully. They could use face-to-face interactions with facility staff to ask which providers fits their schedule and had experience with other cases like theirs. They did not have to spend extra time evaluating whether the chosen online platform was suitable to address their health concern.

DISCUSSION

Attitude

The theory of planned behaviour defines attitude as the tendency to make a decision that is likely to produce positive feelings.[15] Users from the urban site shared that the lack of specialist care demotivates use. These respondents desired to consult with specialists because there was an increased likelihood of being given an immediate diagnosis. Direct consultation with specialists lessened their anxiety of expending time and resources before their concern is addressed. Positive feelings about telemedicine services were associated with a provider's capacity to answer a patient's queries about their health concerns.

Non-users in the urban site believed that the convenience of telemedicine services varied on a 'case-to-case basis'. Telemedicine was considered inconvenient since they cannot fully express themselves in an online setting. Non-users relayed feelings of anxiety when contemplating telemedicine usage. Being physically separated from their providers made them feel that they were outsiders each others' worlds. This perceived mutual exclusion decreased the feeling of trust necessary in building a therapeutic relationship. Participants in the rural and remote site similarly felt anxious about their inability to see their providers' movements and expressions. This sentiment is akin a 2010 article which reported that immersion of a patient's senses into a healthcare setting may make them feel more cared for and attended.[16]

In sum, participants from all sites generally felt that telemedicine is a temporary alternative while in-person consultations were unavailable during the pandemic. Telemedicine was a convenient tool that can accommodate people with busy schedules and those living in localities far from the healthcare centre. However, these perceived benefits were outweighed by hesitancy. Many felt that the therapeutic touch was diminished in an online setup.

Subjective norms

Subjective norms refer to perceived social beliefs or expectations that may influence a participant's decision to use telemedicine services.[15] Access to information is as important as the actual content in influencing decision-making. Participants in all sites agreed that receiving information on telemedicine from representatives showed that healthcare workers were confident in the platform's capacity to support remote consultations.

Effective dissemination of telemedicine announcements depends on the locality's setup.[17] For users and non-users in the urban site, coursing telemedicine-related announcements through offices was considered effective. Participants from the rural and remote sites acknowledged that they were more likely to engage in content promoted through the personal Facebook accounts of CHWs. This can be attributed to the pre-existing level of

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Table 3 Summarised responses on attitudes, subjective norms and perceived behavioural control among telemedicine non-users

Domain	Themes	Urban	Rural	Remote
Attitude	Motivating attributes	► Telemedicine is perceived as an effective tool for managing the number of patients seeking to consultat ► Telemedicine prevents possible exposure to COVID-19 ► Telemedicine is helpful in interpreting lab results	► Telemedicine is perceived as an effective tool for managing the number of patients seeking to consultat ► Telemedicine prevents possible exposure to COVID-19	► Allows provincial workers to perform their jobs ► Allows persons with disabilities and/or senior citizens to access care
	Dettering attributes	► No need for telemedicine because they can easily consult doctors in their social network ► Online consultations feel impersonal ► Perceived it difficult to express themselves in online consultations	► Largely hesitant to use telemedicine services because there is less opportunity to talk with providers	► Open to telemedicine services, but believe that their ailments are not fitting ► Perceived it difficult to express themselves in online consultations
Subjective Norms	Source of information	► Information was mostly obtained through workplace announcements and government officials	► Low exposure to information about telemedicine services; news comes from CHWs	► Information and assistance with accessing telemedicine services is provided by CHWs
	Influencing perceptions	► Hesitant to use telemedicine services because their peers felt that care is better experienced in face-to-face settings	► Non-user peers are generally eager to try telemedicine services but there is some hesitancy due to lack of resources or unfamiliarity	► Non-user peers are generally eager to try telemedicine services but there is some hesitancy due to lack of resources or unfamiliarity
Perceived behavioural control	Reachability of providers	► Concerned that doctors cannot fully perform physical examination of patients in telemedicine platforms	► Feels undersupported by the local health unit if they choose to access telemedicine services	► Perceived that the rural health unit is understaffed and less capable of supporting telemedicine use
	Familiarity	► Feel that they can adapt to any platform. However, they prefer platforms that are straightforward.	► Hesitant to use telemedicine services due to the unfamiliarity of patients and caretakers with gadgets	► Hesitant to use telemedicine services due to the unfamiliarity of patients and caretakers with gadgets
	Resource capacity	► Generally expressed that they won't have difficulty procuring devices or other material resources for teleconsults	► The cost of internet services is prohibitive	► Perceivethat they lack resources such as stable electricity, internet or mobile data connectivity, compatible devices

trust that patients have for CHWs.[18] Aside from healthcare workers, relatives, friends and work colleagues also influenced a patient's decision to access telemedicine care. Positive feedback is more likely to incentivise use.[19] Participants were more optimistic that they can experience good outcomes with telemedicine when someone

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they know has experienced it. The introduction of new healthcare services is optimised when local populations are encouraged to try a service for themselves.[20]

There was no stark difference in subjective norms when we responses were analyzed per age group. Willingness to learn was a greater determinant to telemedicine usage rather than age. Some non-users in the less than 30-year-old group relayed that they would be more likely to explore applications or social media sites that interest them, rather than learning tools that do not satisfy immediate personal goals. Some non-users in the older age groups shared that they would be willing to learn more about telemedicine applications if someone will help them navigate it in the future.

Some non-users in the rural and remote sites expressed that it was their first time knowing that it was possible to consult via text, call or messenger. This shows that an information gap exists in these sites.

Perceived behavioural control

Perceived behaviour control corresponds to the availability of human and material resources that help someone achieve a desired outcome.[15] This also includes the ability of a patient to use available resources at hand.[15] The presence of a user-friendly platform was significant in increasing usage among participants. User-friendliness was predominantly associated with the integration of telemedicine in Facebook—a mainstream platform in the Philippines.[21] Since Facebook was a platform participants used for other socialisation activities; the interface was not challenging to deal with. They can independently consult with providers without having to rely on caretakers.

Participants in the urban site expressed the highest level of behavioural control. Despite this, most users and non-users shared the sentiment that telemedicine provided by UPHS is largely incompatible with their needs. Participants explained that telemedicine services provided did not have sufficient infrastructure to genuinely help patients. It is important to have a system that anticipates a population's needs, complemented by a workforce with compatible skillset. If these are not provided, patients would decline to use the provided service despite necessity.[22]

Users and non-users from rural and remote areas expressed that decreased access to a compatible smartphone, electricity and/or internet load hampered use. Telemedicine access requires adequate economic capital.[23] Participants shared that they repeatedly declined to purchase the resources needed for telemedicine to prioritise basic health needs.

In the remote site, participants expressed that insufficient staffing in the local health unit to support telemedicine services and the frequency of natural calamities decreased their perceived level of control. The absence of healthcare workers was most prominent in the remote site. Healthcare workers are more likely to be present in urban areas due to increased pay and better opportunities for family members.[24] Users in the remote site constantly thought about reverting to face-to-face consultations. They were more certain that a healthcare provider would be available to assist them. These participants also felt that investing in telemedicine-related resources can be potentially counterproductive since signal is usually down during calamities.

Policy recommendations

Augmenting community engagement

To address the lack of familiarity with platforms and gadgets, we recommend stronger engagement down to the community level. This may come in the form of house-to-house or town hall orientations, where residents can learn the skills necessary to navigate the available telemedicine platforms. Patient and caretaker engagement is highly recommended. Pre-existing studies demonstrate that having more stakeholders convinced about the application's benefit increases patients' confidence in the usefulness of telemedicine.[25] Being able to directly communicate with a professional who is knowledgeable about these platforms can allow residents to clarify situational questions related to the application's usage.[26] A trial run of services is strongly encouraged. This allows residents to get accustomed to using telemedicine platforms as part of their healthcare routine before its wide-scale implementation.

Building healthcare worker capacities

CHWs are integral to the healthcare system. They increase the morale toward telemedicine services in the community and connect residents with telemedicine providers.[18] Helping healthcare personnel understand how to manage online platforms produces a more welcoming environment. CHW training also expands the support system of patients during telemedicine consultations. Thus, integration of telemedicine management and use in health personnel provides an avenue for increased utilisation.[5]

Providing infrastructure support

There needs to be greater coordination between the healthcare sector and local government units to ensure adequate infrastructure for communication technology. This is especially necessary for rural and remote settings where there is generally less established infrastructure.[7] Infrastructural support can be facilitated through the identification of areas with strong reception that are easily accessible to residents. These will serve as potential areas for setting up satellite telemedicine sites that residents can access if they wish to avail of remote consultation services. Increased telemedicine access in the home setting could also occur through partnerships with private telecommunication companies that have strong reception in the area. A stronger implementation of EO 127 can facilitate increased access regardless of distance or economic status. These interventions minimise the likelihood that established satellite services become underutilised because cellphones cannot get a signal.

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LIMITATIONS

Different subpopulations have unique healthcare experiences. This study explored geography and age as factors that can influence the decision to use. Other factors such as sex, socioeconomic status and computer literacy can impact individuals' experience of telemedicine. Thus, the results of this study may not be representative of the general Philippine population.

The study participants' attitudes and perspectives towards telemedicine may have also been affected by their recall ability, with perceptions potentially being modified by the passage of time. Future researchers are encouraged to narrow down usage from 3 to 6 months to gain fresh insight.

CONCLUSION

With the advent of the COVID-19 pandemic, telemedicine resurfaces as a potential avenue through which healthcare can remain accessible. It was not seen as a long-term staple for healthcare services, primarily due to technological hesitancy and inadequacy of resources. This suggests that there is work to be done to upscale the usage and acceptance of telemedicine services beyond the pandemic.

This study provides a comparative analysis of the personal and infrastructural factors that may affect telemedicine utilisation among urban, rural and remote localities in LMICs. Analysis of sentiments revealed that the decision to use telemedicine services is largely contextual and socially informed. It adheres to the framework that all domains of behavioural intention affect a patient's decision to use. People make decisions depending on their most recent beliefs about telemedicine, the feedback they receive from people around them, and the perceived availability of resources. Good feedback from trusted individuals and the availability of public infrastructure is likely to produce a positive impact on a person's belief about the service.

This study provides an in-depth analysis of how policymakers can maximise resources to sustain utilisation of telemedicine beyond the pandemic and in anticipation of the universal healthcare implementation in the Philippines. Policy-makers must carefully assess all domains when crafting interventions. The need to create a telemedicine platform that is easy for caretakers and patients to navigate was a distinctive idea expressed by most participants. Healthcare advocates can improve technological infrastructure while making people confident about integrating technology into their healthcare routines.

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REFERENCES

1 Ryu S. History of telemedicine: evolution, context, and transformation. *Healthc Inform Res* 2010;16:65.
2 Noceda AVG, Acierto LMM, Bertiz MCC, *et al*. Patient satisfaction with telemedicine in the Philippines during the COVID-19 pandemic. *Public Glob Health* [Preprint] 2022.
3 Bashshur RL, Shannon GW, Smith BR, *et al*. The empirical foundations of telemedicine interventions for chronic disease management. *Telemed e-Health* 2014;20:769–800.
4 Qiang JK, Marras C. Telemedicine in Parkinson's disease: a patient perspective at a tertiary care centre. *Parkinsonism Relat Disord* 2015;21:525–8.

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5 Macariola AD, Santarin TMC, Villaflor FJM, *et al*. Breaking barriers amid the pandemic: the status of telehealth in Southeast Asia and its potential as a mode of healthcare delivery in the Philippines. *Front Pharmacol* 2021;12:754011.
6 President of the Philippines. Executive order no.127. 2021. Available: https://www.officialgazette.gov.ph/downloads/2021/03mar/20210310-EO-127-RRD.pdf [Accessed 28 Dec 2022].
7 Albert JRG, Quimba FMA, Tabuga AD, *et al*. Expanded data analysis and policy research for national ict household survey report. Philippines Philippine Institute of Development Studies; August 2019.
8 Lau LL, Hung N, Go DJ, *et al*. Knowledge, attitudes and practices of COVID-19 among income-poor households in the Philippines: a cross-sectional study. *J Glob Health* 2020;10:011007.
9 Kohnke A, Cole ML, Bush R. Incorporating UTAUT predictors for understanding home care patients' and clinician's acceptance of healthcare telemedicine equipment. *J Technol Manag Innov* 2014;9:29–41.
10 Sabatello M, Burke TB, McDonald KE, *et al*. Disability, ethics, and health care in the COVID-19 pandemic. *Am J Public Health* 2020;110:1523–7.
11 Gray R, Sanders C. A reflection on the impact of COVID-19 on primary care in the United Kingdom. *J Interprof Care* 2020;34:672–8.
12 Kearon J, Risdon C. The role of primary care in a pandemic: reflections during the COVID-19 pandemic in Canada. *J Prim Care Community Health* 2020;11:2150132720962871.
13 George SM, Hamilton A, Baker R. Pre-experience perceptions about telemedicine among African Americans and Latinos in South Central Los Angeles. *Telemed J E Health* 2009;15:525–30.
14 Bernal-Sundiang N, De Mesa R, MarforiJR, *et al*. Governance in primary care systems: experiences and lessons from urban, rural, and remote settings in the Philippines. *Acta Med Philipp* 2023;57.
15 Godin G, Kok G. The theory of planned behavior: a review of its applications to health-related behaviors. *Am J Health Promot* 1996;11:87–98.
16 Sun R, Blayney DW, Hernandez-Boussard T. Health management via telemedicine: learning from the COVID-19 experience. *J Am Med Inform Assoc* 2021;28:2536–40.
17 Rice T. The hallmark of a doctor': the stethoscope and the making of medical identity. *J Mater Cult* 2010;15:287–301.
18 Campbell C, Gillies P. Conceptualizing 'social capital' for health promotion in small local communities: a micro-qualitative study. *J Community Appl Soc Psychol* 2001;11:329–46.
19 Mallari E, Lasco G, Sayman DJ, *et al*. Connecting communities to primary care: a qualitative study on the roles, motivations and lived experiences of community health workers in the Philippines. *BMC Health Serv Res* 2020;20:860.
20 Jagosh J, Bush PL, Salsberg J, *et al*. A realist evaluation of community-based participatory research: partnership synergy, trust building and related ripple effects. *BMC Public Health* 2015;15:725.
21 Wlodarczyk JR, Wolfswinkel EM, Poh MM, *et al*. Defining best practices for videoconferencing in the era of telemedicine and COVID-19. *J Craniofac Surg* 2020;31:e658–60.
22 John RBA, Lucila OB. Personality traits as predictors of facebook use. *Int J Psychol Couns* 2016;8:45–52.
23 Subramanian U, Hopp F, Lowery J, *et al*. Research in home-care telemedicine: challenges in patient recruitment. *Telemed J E Health* 2004;10:155–61.
24 Nikolian VC, Williams AM, Jacobs BN, *et al*. Pilot study to evaluate the safety, feasibility, and financial implications of a postoperative telemedicine program. *Ann Surg* 2018;268:700–7.
25 Graven LJ, Glueckauf RL, Regal RA, *et al*. Telehealth interventions for family caregivers of persons with chronic health conditions: a systematic review of randomized controlled trials. *Int J Telemed Appl* 2021;2021:3518050.
26 Wilson NW, Couper ID, De Vries E, *et al*. A critical review of interventions to redress the inequitable distribution of healthcare professionals to rural and remote areas. *Rural Remote Health* 2009;9:1060.

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